Machine and Deep Learning Lab

EXPERIMENT NO: 6

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Experiment:

Implement K-Means and Hierarchical Clustering on Sample Dataset

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Aim

To implement and analyze **K-Means Clustering** and **Hierarchical Clustering** on the Iris dataset, emphasizing unsupervised learning and validating cluster structure using multiple evaluation metrics.

Dataset Source

Dataset: Iris Dataset
Source: https://scikit-learn.org/stable/datasets/toy_dataset.html#iris-dataset

Access Method:

```
from sklearn.datasets import load_iris
```

Dataset Description

The Iris dataset contains **150 samples** of iris flowers divided into **three species**.

Feature	Description	Type
Sepal Length	Length of sepal	Numeric
Sepal Width	Width of sepal	Numeric
Petal Length	Length of petal	Numeric
Petal Width	Width of petal	Numeric

For Clustering:

- All 4 features are used
- Labels are **not used during training**
- Labels are used only for **Adjusted Rand Index (ARI) evaluation**

Theory

K-Means Clustering

K-Means partitions data into **K clusters** by minimizing the within-cluster sum of squares.

Objective Function:

$$J = \sum_{j=1}^{K} \sum_{i=1}^{n} \|x_i^j - c_j\|^2$$

Where:

- KK = number of clusters
- cjcj = centroid of cluster jj
- xijxij = data point in cluster jj

Hierarchical Clustering

Hierarchical clustering builds nested clusters using a **bottom-up (agglomerative) approach**.

Ward Linkage:

$$d(p, q) = \sum (q_i - p_i)^2$$

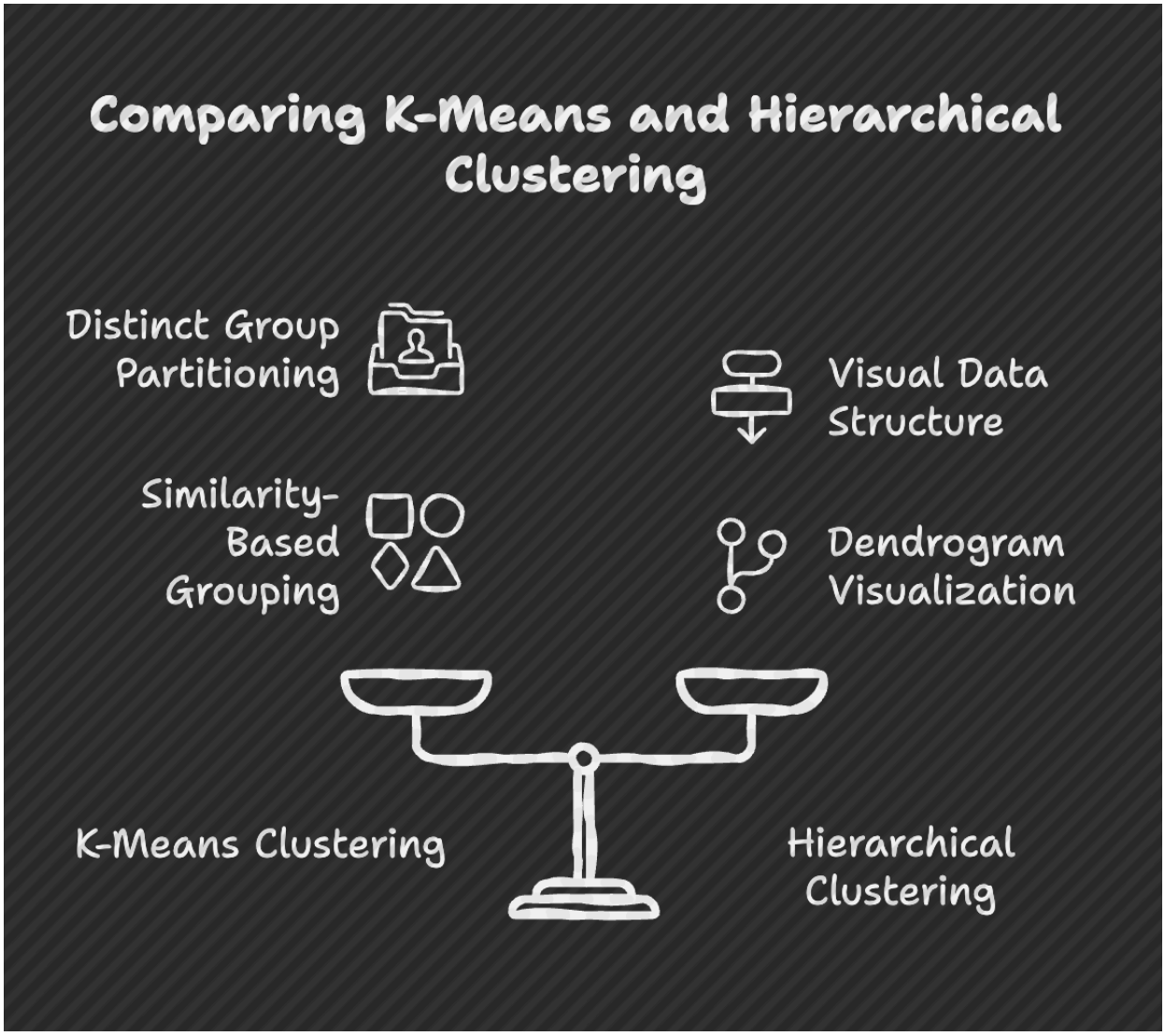
Ward's method minimizes the increase in within-cluster variance.

Methodology / Workflow

Step-by-Step Procedure:

1. Import required libraries
2. Load Iris dataset
3. Feature scaling using StandardScaler
4. Apply K-Means clustering
5. Apply Hierarchical clustering
6. Perform hyperparameter tuning
7. Plot dendrogram
8. Evaluate clusters using performance metrics
9. Compare results

Workflow Diagram:



Dataset → Feature Scaling → Clustering → Hyperparameter Tuning → Evaluation

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Performance Evaluation Metrics

Silhouette Score

- Measures cluster separation
- Range: **-1 to 1**
- Higher value → Better clustering

Davies-Bouldin Index (DBI)

- Measures cluster compactness and separation
- Lower value → Better clustering

Adjusted Rand Index (ARI)

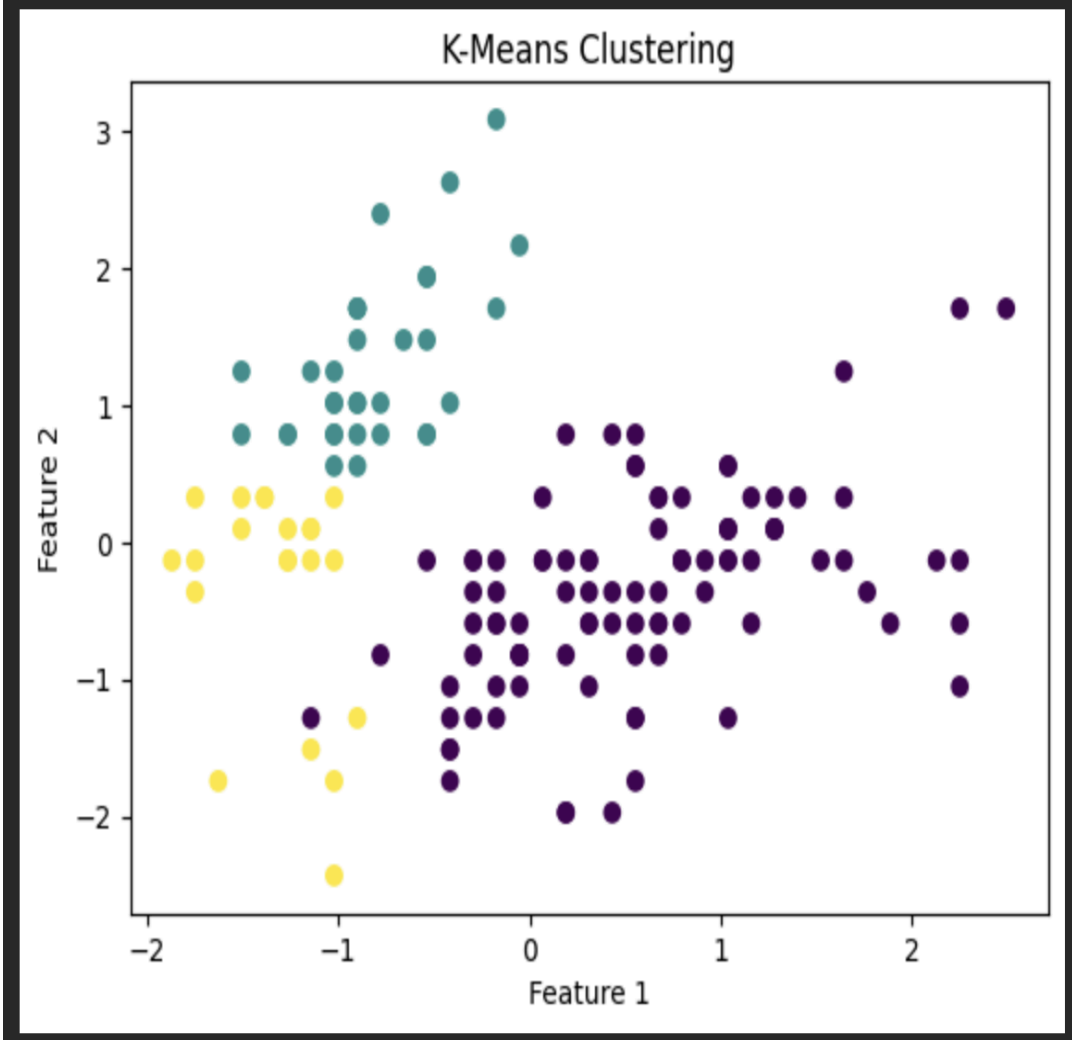
- Compares clustering with true labels
- Range: **-1 to 1**
- Value close to 1 → High similarity

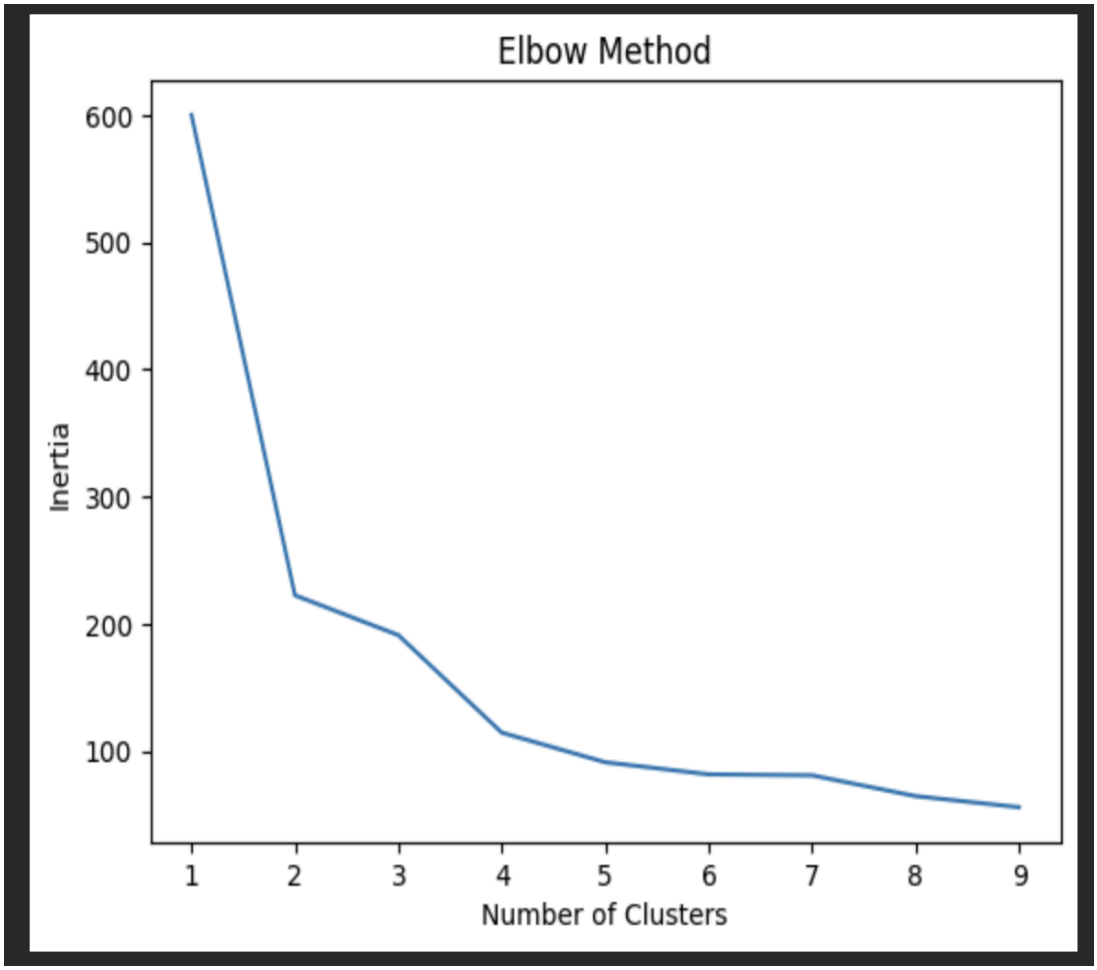
Hyperparameter Tuning

K-Means Tuned Parameters:

- Number of clusters (K = 2 to 6)
- n_init (number of initializations)

```
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=clusters)
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("K-Means Clustering")
plt.show()
```





Impact:

- Optimal K gives best silhouette score
- Higher n_init avoids local minima and improves clustering quality

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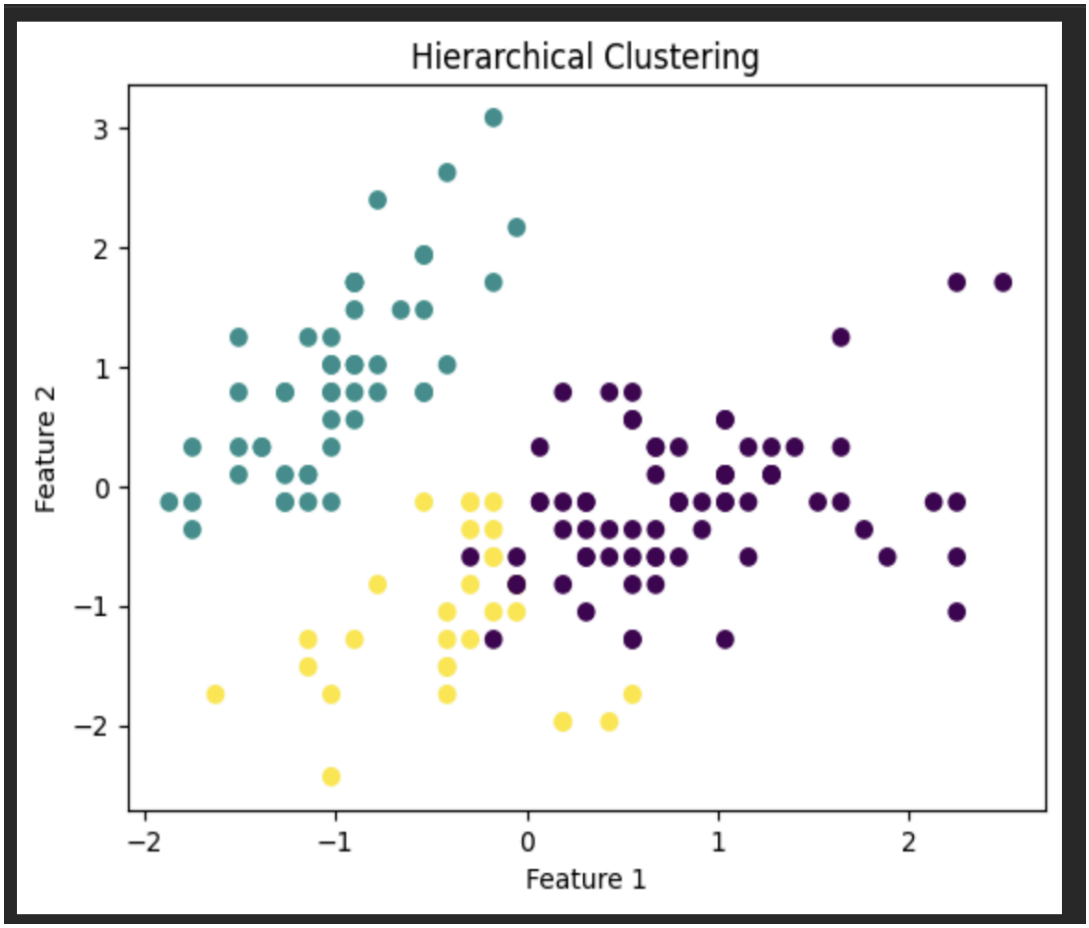
Hierarchical Clustering Tuned Parameters:

- Linkage method
  - ward
  - average
  - complete

Impact:

- Ward linkage produces compact and well-separated clusters
- Different linkage methods change cluster structure

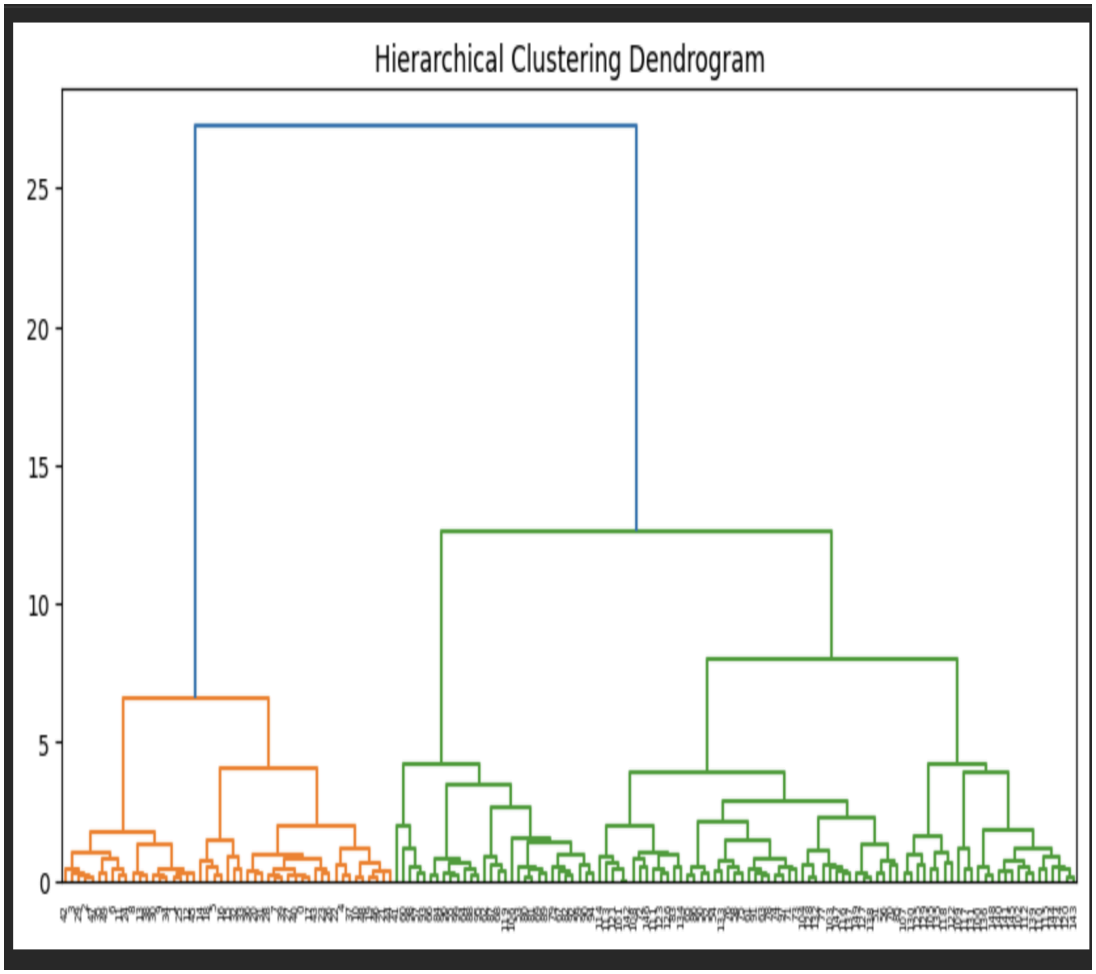


Dendrogram

[Space for Dendrogram Figure]

Used to:

- Visualize hierarchical cluster formation
- Determine optimal number of clusters



Performance Analysis

Typical observations for Iris dataset:

- **Silhouette Score ≈ 0.55 – 0.65** → Well-separated clusters
- **Davies-Bouldin Index < 0.6** → Compact and distinct clusters
- **ARI close to 1** → Strong alignment with actual species

Since Iris naturally contains **3 clusters**, the clustering algorithms achieve:

- High ARI
- Good silhouette score
- Clear cluster structure

Technical Justification

- Feature scaling ensured uniform distance computation
- Increasing **n_init** improved global optimization in K-Means
- Ward linkage preserved compact cluster structure
- Hyperparameter tuning improved cluster separation and compactness

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Important Difference from Mall Customer Dataset

Mall Dataset	Iris Dataset
2 features	4 features
Easy 2D visualization	Requires PCA for visualization
No true labels for validation	ARI can be computed

Google Colab Link

EXPERIMENT 6.ipynb

Result

K-Means and Hierarchical clustering successfully identified the inherent grouping in the Iris dataset.
After hyperparameter tuning, both algorithms produced more compact and well-separated clusters with improved evaluation metric values.

Conclusion

This experiment demonstrates that K-Means and Hierarchical clustering effectively discover the natural structure of the Iris dataset.
Hyperparameter tuning significantly improves cluster quality and consistency.
It also validates the importance of feature scaling and proper parameter selection in unsupervised learning.